

appropriate family for a problem, then the use of the more general families we have studied is foolish and a waste of both machine time and insight. Since the appropriate family to use is so difficult to discuss in the general case, we shall confine the chapter to approximating particular cases and assume that the reader realizes that in other cases the same techniques apply and should be used.

Much of mathematics involves the art of studying the behavior of functions in the neighborhood of a singularity, and we cannot in this book go into this important topic—we must simply assume that it is known, especially the difficult art of handling “movable singularities.”

In mathematics the singularities are usually subtracted out, but in computing it is usually better to arrange for the behavior in the neighborhood of the singularity to be written as a product of the known general behavior times an unknown function that will, if the choice is good, remain finite and vary slowly in the region of interest. We must, therefore, resist using many of the standard tricks taught in mathematics and learn to proceed somewhat differently, if we are to avoid having to deal with differences of large numbers and the consequent heavy cancellations.

42.2 SOME EXAMPLES OF INTEGRALS WITH SINGULARITIES

As a first example, consider the integration of

$$I = \int_0^{\pi/2} \ln(\sin x) dx$$

If we write it as

$$I = \int_0^{\pi/2} \left[\ln x + \ln\left(\frac{\sin x}{x}\right) \right] dx$$

we see that it can be written as a piece that we can integrate analytically and which contains the singularity plus a second piece which we can hope to do numerically since the integrand is now well behaved

$$I = \frac{\pi}{2} \left(\ln \frac{\pi}{2} - 1 \right) + \int_0^{\pi/2} \ln \left(\frac{\sin x}{x} \right) dx$$

Using $h = \pi/4$ and Simpson's formula, we get

$$\begin{aligned} I &= -0.862 + \frac{\pi}{12} (0 - 0.420 - 0.045) \\ &= -0.862 - 0.228 = -1.090 \\ \left(\text{Known answer} &= -\frac{\pi}{2} \ln 2 = -1.089 \cdots \right) \end{aligned}$$

Here we have used a mixture of subtracting out the singularity and writing it as a product.

Second, it is often possible to find a transformation that eliminates the singularity. Thus in the problem in Sec. 9.10

$$g(y) = \frac{d}{dy} \int_0^y \frac{f(x)}{\sqrt{y-x}} dx$$

the transformation

$$x = y \sin^2 \theta$$

removed the singularity. This again falls in the domain of formal mathematical manipulation and outside that of numerical methods.

As a third example, consider

$$\begin{aligned} f(x) &= \int_a^x e^{x^2} dx = \int_a^x \frac{1}{2x} (2xe^{x^2}) dx \quad a > 0 \\ &= \frac{e^{x^2}}{2x} \Big|_a^x + \int_a^x \frac{e^{x^2}}{2x^2} dx \end{aligned}$$

so that the asymptotic expansion of $f(x)$ is

$$f(x) \simeq \frac{e^{x^2}}{2x} + \cdots$$

Near $x = 0$ this form is not useful and we are led to try (now $a = 0$)

$$f(x) = \int_0^x e^{x^2} dx = e^{x^2} D(x)$$

It is a common trick to get a differential equation from an integral, and so we differentiate to get

$$e^{x^2} D'(x) + 2xe^{x^2} D(x) = e^{x^2}$$

or

$$D'(x) + 2xD(x) = 1$$

$$D(0) = 0$$

We now have a simple differential equation and know that as $x \rightarrow \infty$, $D(x) \rightarrow 1/(2x)$. This suggests that the function $D(x)$ (for the whole range $0 \leq x \leq \infty$), known as Dawson's integral, is more suitable for computing than the original function.

As a fourth example, consider Bode's integral for the phase in terms of the amplitude $A(\omega)$:

$$\frac{\omega_0}{\pi} \int_{-\infty}^{\infty} \frac{A(\omega)}{\omega^2 - \omega_0^2} d\omega$$

Here we have singularities at

$$\omega = \pm \omega_0$$

A numerical computation from experimental data used the trapezoid rule for the intervals outside of the singularities, and near the singularities the function $A(\omega)$ was approximated by a cubic through four adjacent, centrally located values. This led to integrals of the form

$$\int_{\omega_0-h/2}^{\omega_0+h/2} \left[\frac{a + b(\omega - \omega_0) + c(\omega - \omega_0)^2 + d(\omega - \omega_0)^3}{(\omega - \omega_0)(\omega + \omega_0)} \right] d\omega$$

which can be handled analytically.

The use of the structure of the singularity has been illustrated to some extent by these examples, and further examples occur in the next section.

PROBLEMS

42.2.1 Writing

$$I = \int_{-1}^1 \frac{dx}{\sqrt{1-x^4}} = \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} \frac{1}{\sqrt{1+x^2}} dx$$

apply Chebyshev integration.

42.2.2 In Prob. 42.2.1, set $x = \cos \theta$ and integrate numerically. Compare to Prob. 42.2.1.

42.3 A SINGULARITY IN A LINEAR DIFFERENTIAL EQUATION

It occasionally happens that a numerical solution is required for a linear differential equation which has a singularity¹ in the range of integration. One way of doing this, which was first suggested by Prof. J. W. Tukey, is the following, and it again illustrates the technique of using the structure of the singularity in a multiplicative fashion.

Suppose we are given

$$y'' + P(x)y = 0 \quad \begin{aligned} y(x_0) &= A \\ y'(x_0) &= B \end{aligned}$$

¹ By singularity we mean that the function does not behave as a power series about the point.

where $P(x)$ has a singularity in the range of integration, say at $x = 0$. We choose as a comparison equation

$$u'' + Q(x)u = 0$$

which has the known solutions $u_1(x)$ and $u_2(x)$ and has the same kind of singularity in $Q(x)$ at the same place as $P(x)$. We shall later examine how this can be done.

We next assume that

$$y(x) = \alpha(x)u_1(x) + \beta(x)u_2(x)$$

where $\alpha(x)$ and $\beta(x)$ are functions to be determined but which we expect will vary smoothly. The method now goes along the same lines as the classical method of "variation of parameters." Having introduced two unknown functions $\alpha(x)$ and $\beta(x)$, we may require that $y(x)$ satisfy both the differential equation and one more condition. We choose this second condition as

$$\alpha'u_1 + \beta'u_2 = 0$$

From this we have

$$y = \alpha u_1 + \beta u_2$$

$$y' = \alpha u_1' + \beta u_2'$$

$$y'' = \alpha u_1'' + \alpha' u_1' + \beta u_2'' + \beta' u_2'$$

Putting these in the original equation, we get

$$\alpha(u_1'' + Pu_1) + \beta(u_2'' + Pu_2) + \alpha' u_1' + \beta' u_2' = 0$$

But using the fact that u_1 and u_2 are solutions of the comparison equation, we have

$$u_1'' + Pu_1 \equiv (P - Q)u_1$$

$$u_2'' + Pu_2 \equiv (P - Q)u_2$$

Thus the differential equation becomes

$$\alpha' u_1' + \beta' u_2' = (Q - P)(\alpha u_1 + \beta u_2) = (Q - P)y$$

Solving this with the definition of $y(x)$, we get the two equations

$$\alpha'(u_1 u_2' - u_2 u_1') = -u_2(Q - P)y$$

$$\beta'(u_1 u_2' - u_2 u_1') = u_1(Q - P)y$$

But $u_1 u_2' - u_2 u_1'$ is the Wronskian of u_1 and u_2 , and for this case (no y' term) it is a constant W_0 , where

$$W_0 = u_1(x_0)u_2'(x_0) - u_2(x_0)u_1'(x_0)$$

W_0 is readily determined from the known solutions $u_1(x)$ and $u_2(x)$.

We thus have to solve

$$\begin{aligned}\alpha' &= -\frac{u_2(Q-P)y}{W_0} \\ \beta' &= \frac{u_1(Q-P)y}{W_0} \quad \begin{aligned} y(x_0) &= A \\ y'(x_0) &= B \end{aligned} \\ y &= \alpha u_1 + \beta u_2\end{aligned}$$

The problem is how we can choose a $Q(x)$ so that the factor $Q - P$ vanishes sufficiently strongly at the singularity to cover the peculiar behavior of the products $u_2 y$ and $u_1 y$ (or, equivalently, the products u_2^2 , $u_1 u_2$, and u_1^2). If this can be done, then the equations determining $\alpha(x)$ and $\beta(x)$ will have a much less peculiar behavior at the point where there was a singularity.

We now show how $u_1(x)$ and $u_2(x)$ may be chosen in one particular example. Suppose that

$$P(x) = \frac{a_{-1}}{x} + a_0 + a_1 x + \cdots \quad a_{-1} \neq 0$$

Thus we know that if $y(0) \neq 0$, then $y''(0)$ is infinite, and it is apparent that we could not approximate the solution by a polynomial with any fidelity near $x = 0$. We try

$$u_1 = \frac{x}{1 + ax + bx^2} \equiv \frac{x}{D(x)} \equiv xD^{-1}(x)$$

Then

$$\begin{aligned}u_1'' &= [2(D^{-1})' + x(D^{-1})''] \left(\frac{D}{x} u_1 \right) \quad \frac{D}{x} u_1 \equiv 1 \\ &= \frac{x[2(D')^2 - DD''] - 2DD'}{xD^2} u_1\end{aligned}$$

But

$$\begin{aligned}D &= 1 + ax + bx^2 \\ D' &= a + 2bx \\ D'' &= 2b\end{aligned}$$

Hence for small x

$$\begin{aligned}u_1'' &= \frac{-2a - 6bx + 2b^2 x^2}{xD^2} u_1 = \frac{-2a - 6bx + 2b^2 x^2}{x[1 + 2ax + (a^2 + 2b)x^2 + \cdots]} u_1 \\ &= -2 \left[\frac{a}{x} + (3b - 2a^2) + \cdots \right] u_1\end{aligned}$$

Thus

$$Q = -\frac{2a}{x} + (-6b + 4a^2) + (\cdots)x + \cdots$$

and we want to choose (comparing Q and P)

$$\begin{aligned} -2a &= a_{-1} & \text{or} & & a &= \frac{a_{-1}}{2} \\ -6b + 4a^2 &= a_0 & & & b &= \frac{a_{-1}^2 - a_0}{6} \end{aligned}$$

With this choice of a and b in our trial function $Q(x)$, we have from the structure of $P(x)$

$$Q - P \simeq Cx + \cdots$$

and $Q - P$ vanishes sufficiently rapidly. We now determine u_2 in the usual fashion.

$$\begin{aligned} u_2(x) &= u_1(x) \int^x \frac{d\theta}{u_1^2(\theta)} \\ &= u_1(x) \int^x \frac{1 + 2a\theta + (a^2 + 2b)\theta^2 + 2ab\theta^3 + b^2\theta^4}{\theta^2} d\theta \\ &= u_1(x) \left[-\frac{1}{x} + 2a \ln|x| + (a^2 + 2b)x + abx^2 + \frac{b^2x^3}{3} \right] \end{aligned}$$

and the rest is easy. We have $u_1(x)$ with no singularity at $x = 0$, and $u_2(0) = -1$. We also have a $u_2(x)$ whose second derivative is infinite at $x = 0$. The products

$$(Q - P)u_1y \quad \text{and} \quad (Q - P)u_2y$$

vanish at $x = 0$, allowing the smooth integration of the differential equations that determine $\alpha(x)$ and $\beta(x)$, since the peculiar behavior is contained in the u_2 and is partially masked by the zero of $Q - P$.

In this example we again see the advantage of handling the singularities by first examining the dominant behavior of the solution near the singularity and then using a known function to approximate this behavior.

42.4 GENERAL REMARKS

We have given some examples of how to handle singularities in a few specific cases. We have no detailed rules for the general case, especially for singularities in nonlinear problems. Much depends on the mathematical ability of the person planning the solution.

With modern floating-point machines it is sometimes possible to integrate close enough to the singularity so that an analytic approximation can be fitted and used to get over the singularity. Once past the singularity, numerical values can be computed from the analytic expression in order to resume the computation. However, this should be used only when better methods fail, as the accuracy is often hard to estimate and control.

When the singularity is a so-called “movable singularity,” which can occur in nonlinear problems, then it is particularly important to be careful how the numerical data are used to fit the function approximating the singularity. Yet such things can be and have been done successfully; it is a matter of courage and careful computation.

Experience shows that frequently the singularities were not in the original physical problem but got there by some standard mathematical transformation. Thus there is a high probability that a transformation can be found which removes the singularity, and that you are undoing what the mathematician did before you got the problem. It is therefore highly recommended that if a problem turns up with a singularity in it, then a careful examination of the source should be made to see how it arose. Occasionally it is in the original formulation, but an equally satisfactory approximation theory without the singularity can be made—often more satisfactory from the computing point of view. In the author’s experience once after solving, with great labor, a problem having a singularity in it, the proposer asked if it could be done with some viscous terms in it—terms that effectively eliminated the trouble! What appears to the proposer to be a more complicated and more realistic model often is easier for the computer expert to handle.

43.1 INTRODUCTION

Simulations are usually undertaken with the goal of ultimately optimizing something about the behavior of the system. Thus the usual steps in a simulation are:

- 1 Construct the model of the situation (or process)
- 2 Decide on the objective function that combines *all* the various goals into one single numerical value
- 3 Optimize

Each of these three steps is much harder than is usually realized when a study is first proposed.

First, the situation to be simulated is usually not as well defined as was supposed. Many parts have not been thought through, and some parts do not have their proper physical constants for complete specification, since usually the simulation is supposed to help in the design. In the ideal situation the simulation grows into the design, and that in turn flows into the evaluation of the system; it is wrong to separate the three phases.

Second, the so-called “objective function,” which is supposed to combine all the various goals and desirable properties of the system into one single real

number, is not known at the start. The question of “trade-offs” of this for that has not been thought through, and will probably not be until some of the results are available so that people can begin to think about those that will matter in the final design. At the beginning of the study the idea of a single number measuring the quality of the whole complex system does violence to the engineer’s feelings of what is important. Yet, in the final analysis, somehow the noncomparable aspects must be combined and a single decision reached as to which one of the possible designs to use.

Third, while it is the simulation that takes so much machine time, both in the programming and in the actual running, it is the optimization process which often controls the success or failure. Inefficient searches of the space of possible designs can require more time and energy than are available. Thus the part that tends to get neglected in the original planning often dominates the question of success or failure.

It is necessary, therefore, to look at the optimization process. There are, broadly speaking, two situations. First, the classical one that arose in agricultural experimentation where, because of the length of time to do one experiment, many cases must be run at the same time (in parallel). From them a conclusion is drawn. Second, modern situations that generally allow sequential decisions; one case is run, and then a partial analysis of the results is made which is used to help select the next trial. It should be intuitively clear that the sequential process is generally very much more powerful in arriving at the optimum in a few steps than is the parallel case. These two approaches have given rise to two distinct fields; the parallel one that arose from agricultural experiments gave rise to the field called “design of experiments,” while the sequential approach has given rise to the more recent “sequential design” and the still rapidly developing field of optimization.

The problem of finding the optimum has a number of traps for the unwary person who believes everything he reads. For example, consider the two situations in Fig. 43.1.1. In the first, with a broad minimum, the mathematician will say “it is ill-conditioned and difficult to find,” while in the second he will say “it is well-conditioned and easy to find.” The thoughtful engineer who is going to use the results will soon realize that in the first case he can safely proceed, because small changes in the position of the x value, or in the mathematical model, or again in the physical system that is ultimately constructed, will make small differences in the quality (y) of the performance. On the other hand, for the sharp minimum the opposite applies, and small changes will greatly affect the behavior (y) of the resulting system. Thus the mathematician’s view is almost diametrically opposite to that of the engineer’s, and it is the engineer who is often right in the matter.

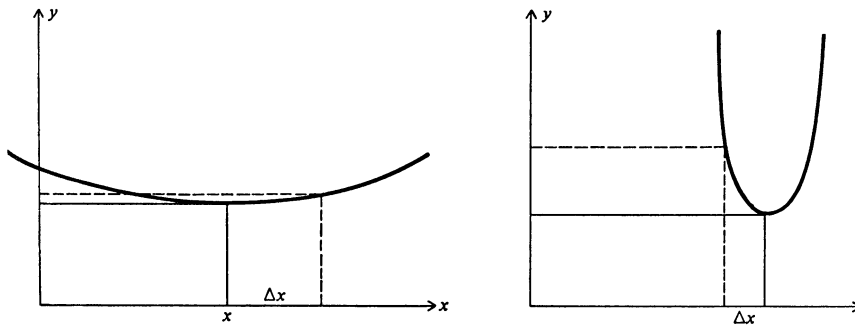


FIGURE 43.1.1

43.2 REVIEW OF CALCULUS RESULTS

Optimization is a standard topic in the calculus, but it is necessary to recall some vital details that tend to be forgotten. The main method for functions of a single variable is to obtain the first derivative, set it equal to zero, and then find the roots of this equation. These roots give the positions of the extremes (maxima,

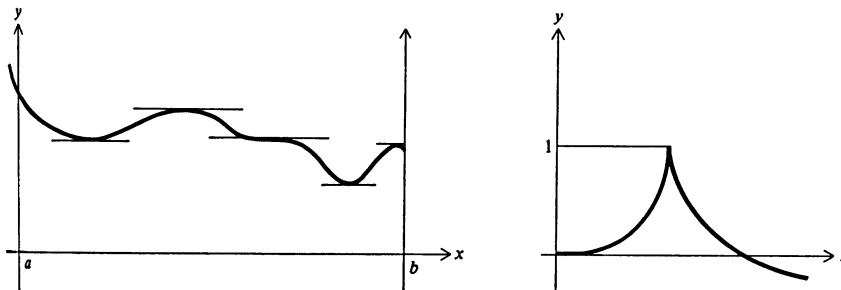
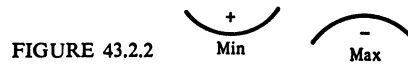


FIGURE 43.2.1

minima, and horizontal tangents). In the textbook examples these steps are usually fairly easy to do, but in practice the real zeros (Chap. 4) can be difficult to find.

What is usually glossed over are the assumptions that not only the function, but also the derivative, is continuous—that there are no cusps—and that the extreme value does not occur on the boundary. Both are significant in practice.

The test for a maximum is that the second derivative at the extreme point be negative, while for a minimum the second derivative must be positive (see Fig. 43.2.2). If the second derivative is zero at the point, then higher derivatives must be examined. If the first nonzero derivative at the point is of odd order, then it is an inflection point, while if it is of even order, then a sign test like that for the second derivative applies.



For functions of two variables the matter is much more complex. The usual test for two variables is

$$\frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 \begin{cases} > 0 & \frac{\partial^2 f}{\partial x^2} \begin{cases} < 0 & \text{maximum} \\ > 0 & \text{minimum} \end{cases} \\ = 0 & \text{undecided} \\ < 0 & \text{neither max. nor min.} \end{cases}$$

(again recall that this applies only on the inside of the region and that the boundaries require special attention). The textbooks rarely go beyond two variables.

The classic example of the peculiar behavior of a function of two variables near what seems to be a minimum is

$$\begin{aligned} z &= (y - x^2)(y - 2x^2) \\ &= y^2 - 3x^2y + 2x^4 \end{aligned}$$

At the origin

$$\begin{aligned} \frac{\partial z}{\partial x} &= -6xy + 8x^3 = 0 \\ \frac{\partial z}{\partial y} &= 2y - 3x^2 = 0 \\ \frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 &= 0 \end{aligned}$$

Now consider planes that pass through the z axis, $y = \lambda x$. These planes cut the surface in the curves

$$z = \lambda^2 x^2 - 3\lambda x^3 + 2x^4$$

and for them

$$\frac{dz}{dx} = 0$$

$$\frac{d^2z}{dx^2} = 2\lambda^2 > 0 \quad \text{a minimum at } x = 0, y = 0 \text{ in the plane}$$

hence we have a minimum in each plane. The two extreme cases (planes that also contain the x axis or the y axis) also show a minimum at the origin. Thus we are inclined to think that the origin is a true minimum. See Fig. 43.2.3. However, in every neighborhood of the origin the function takes on negative values! If functions of two variables can be this tricky, consider those of many more variables!

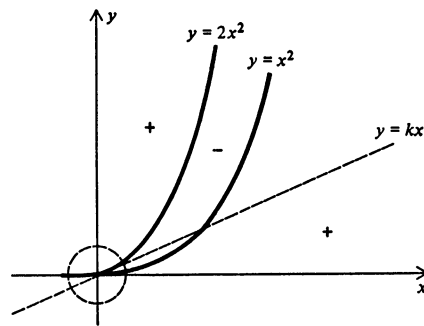


FIGURE 43.2.3

It is also worth noting that many apparently simple problems have no mathematically defined minimum value. For example, suppose that there are two towns A and B , with B due east of A , and suppose that you are required to depart from A in a northerly direction and travel the minimum path to B . See Fig. 43.2.4. Evidently in a mathematical sense this problem is not really properly stated, though at first glance it might appear to be. In practical situations the foolishness is generally better disguised, but may be there just the same to trap the unwary.

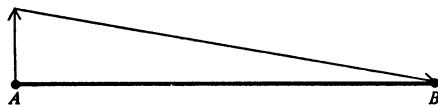


FIGURE 43.2.4

43.3 LAGRANGE MULTIPLIERS

Frequently the problem we are given to minimize has n variables, but there are k constraints among the variables. Thus we have to minimize

$$y = f(x_1, x_2, \dots, x_n)$$

when

$$\begin{cases} \phi_1(x_1, x_2, \dots, x_n) = 0 \\ \dots\dots\dots \\ \phi_k(x_1, x_2, \dots, x_n) = 0 \end{cases} \quad k < n$$

For example, the values of the points $(x_1, x_2, \dots, x_n)$ may be required to lie on the surface of a sphere

$$\phi = x_1^2 + x_2^2 + \dots + x_n^2 - a^2 = 0$$

In principle the k constraint equations can be solved for k of the variables x_i , and these can be used to eliminate these variables from the original function $y = f(x_1, x_2, \dots, x_n)$. In practice this is apt to be impossible to carry out, and a trick known as the method of Lagrange multipliers can be used instead. We will illustrate it with one constraint. Using differentials, we have

$$dy = \frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f}{\partial x_n} \Delta x_n = 0$$

and

$$0 = \frac{\partial \phi}{\partial x_1} \Delta x_1 + \frac{\partial \phi}{\partial x_2} \Delta x_2 + \dots + \frac{\partial \phi}{\partial x_n} \Delta x_n = 0$$

Using λ times the second equation added to the first, we get

$$\left(\frac{\partial f}{\partial x_1} + \lambda \frac{\partial \phi}{\partial x_1} \right) \Delta x_1 + \dots + \left(\frac{\partial f}{\partial x_n} + \lambda \frac{\partial \phi}{\partial x_n} \right) \Delta x_n = 0$$

If each of these parentheses vanishes, then for any variation of the Δx_i we will have a horizontal tangent; thus the n equations

$$\frac{\partial f}{\partial x_i} + \lambda \frac{\partial \phi}{\partial x_i} = 0 \quad i = 1, 2, \dots, n$$

plus the original constraint

$$\phi(x_1, x_2, \dots, x_n) = 0$$

gives us $n + 1$ equations in the $n + 1$ variables $x_1, x_2, \dots, x_n$ and λ . These equations define the solution we are looking for. Thus in place of the original function we wanted to minimize we now consider the function

$$L \equiv f + \lambda \phi$$

and optimize this as a function of the x_i with λ as an additional variable.

As an illustration of the Lagrange multiplier method, consider the problem of finding the maximum rectangular block that will fit inside the ellipsoid

$$\phi = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0$$

The volume to be maximized is

$$V = 8xyz$$

subject to the constraint of the point x, y, z lying on the ellipsoid. We therefore consider the function

$$L = V + \lambda\phi$$

and have to solve the equations

$$\frac{\partial L}{\partial x} = 8yz + \lambda \frac{2x}{a^2} = 0$$

$$\frac{\partial L}{\partial y} = 8xz + \lambda \frac{2y}{b^2} = 0$$

$$\frac{\partial L}{\partial z} = 8xy + \lambda \frac{2z}{c^2} = 0$$

together with the original constraint, the ellipsoid. To solve these equations, multiply the first equation by x , the second by y , and the third by z , and add to get

$$24xyz + 2\lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right) = 0$$

or

$$3V + 2\lambda = 0$$

Again multiply the top equation by x to get

$$8xyz + 2\lambda \frac{x^2}{a^2} = 0$$

or

$$V + 2\lambda \frac{x^2}{a^2} = 0$$

Using this with the above equation connecting V and λ , we get

$$V - 3V \frac{x^2}{a^2} = 0$$

Excluding $V = 0$ as a minimum, we get

$$x^2 = \frac{a^2}{3}$$

$$x = \frac{a}{\sqrt{3}}$$

By symmetry we also get

$$y = \frac{b}{\sqrt{3}}$$

$$z = \frac{c}{\sqrt{3}}$$

and the maximum volume is

$$V = \frac{8abc}{3\sqrt{3}}$$

To generalize the Lagrange multiplier method when there are k constraints, we merely use k Lagrange multipliers λ_i and proceed accordingly.

PROBLEMS

43.3.3 Find the minimum distance from the plane $Ax + By + Cz - D = 0$ to the origin.

$$\text{Ans. } \frac{|D|}{\sqrt{A^2 + B^2 + C^2}}$$

43.4 THE CURSE OF DIMENSION

We have looked at, and will continue to use, problems in two independent variables, but in practice the number of parameters in the problem being optimized may run much higher, and it is necessary to discuss the difference between problems in a few dimensions and those in many. The difference is known as “the curse of dimension,” as the following discussion will show.

Consider a unit cube in n dimensions. It will have volume $1^n = 1$. Now consider all the volume in the cube that is within a distance of ε from the surface. This volume is the difference between two cubes, the original cube of side 1 and the inner cube of side $1 - 2\varepsilon$. This difference is

$$1 - (1 - 2\varepsilon)^n$$

and we see that for a fixed ε , when we let n get larger and larger, the difference approaches 1. Thus in a high-dimension space almost all the volume of a cube is near the surface!

For a sphere we have a similar result. The volume is proportional to the n th power of the radius.

$$V = \frac{\pi^{n/2} r^n}{\Gamma(n/2 + 1)}$$

Thus as we search high-dimension spaces for optimal values, we find that almost all the volume is on the surface and that what in low-dimension spaces was a nuisance is now a necessity: we must pay close attention to the values on the surface of the region, since we can expect that frequently the optimum will be there.

In many cases that are casually talked about, the volume of the high-dimension space is so great that it is impracticable to explore it in any detail at all. Problems with 50 and 100 parameters are hopeless *unless* a great deal more is known about where to look. For example, if it is known that there is only one local maximum, then it is a reasonable problem. Usually only local extreme values can be found, with a high probability of significantly higher values being somewhere in the region where you have never looked. The methods we shall give are practical only for comparatively few parameter problems, or else in controlled situations where few extremes can occur.

43.5 THE GRADIENT

The methods of the calculus attempt to find the extreme points by direct methods in that they produce the answer without looking at other places. These methods fail when there is no nice, simple, explicit formula for the objective function which measures the goodness of the trial.

We now turn to methods which resemble hill climbing or valley descending. They picture the surface generated by the objective function as a topographic map with the level lines drawn as shown in Fig. 43.5.1. The level lines are defined by

$$f(x_1, x_2, \dots, x_n) = \text{constant}$$

where of course $f(x_1, \dots, x_n)$ is the objective function.

Since the maxima of f are the minima of $-f$, we are not limiting ourselves if we search for minima. The method consists, crudely, in taking the next step so as to go downhill rapidly—it follows the local topography and ignores any attempt to see the surface as a whole. The main weakness of this method of finding the minima of a function by valley descending is that it is apt to miss very narrow, deep, pits, such as a mine opening. If the search of the map does not

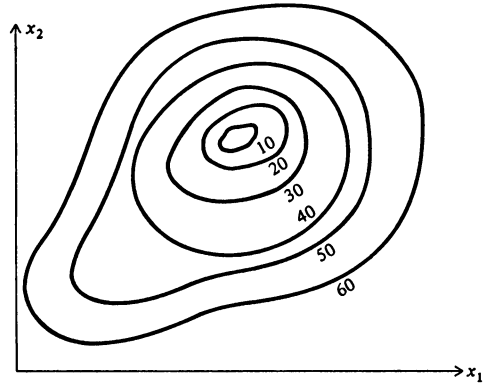


FIGURE 43.5.1

by chance run into the opening, then it will not be found by the method of searching the surface step by step. What the method does find is the minimum of the local (smooth) valley that you start in.

The level curves (surfaces and hypersurfaces if $n > 2$) are defined by equating the objective function to the appropriate constant value. If we differentiate this function, holding the value constant (along the level curve), we get

$$\frac{dy}{ds} = 0 = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial s} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial s} + \cdots + \frac{\partial f}{\partial x_n} \frac{\partial x_n}{\partial s}$$

where s is the arc length along the level curve. The n functions

$$\frac{\partial x_j}{\partial s} \quad j = 1, 2, \dots, n$$

form a vector which gives the direction cosines of the level curve. It follows that the functions

$$\frac{\partial f}{\partial x_j} \quad j = 1, 2, \dots, n$$

also define a vector of direction numbers. We shall indicate a vector by boldface type as

$$\left(\frac{\partial \mathbf{f}}{\partial \mathbf{x}_j} \right)$$

The original direction cosines $\partial x_j / \partial s$ define the local direction of the level curve. The condition for orthogonality of two vectors is that the sum of the

products of the corresponding components be zero (see Sec. 26.1). Thus the above equation, which we found by differentiating the surface along the level curves, shows that the two vectors are perpendicular. One is the direction numbers of the level curve, so that the other

$$\left(\frac{\partial f}{\partial x_j} \right) \equiv \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

which is called *the gradient*, is perpendicular to the level curve. It is easy to see that the negative gradient is the path of steepest descent (locally). The gradient changes its direction from place to place, but if we follow the local gradient downward, we will get to the (local) minimum.

The path of steepest descent is not necessarily the shortest path to the minimum. Consider a meandering river. Except for the inertia of the flowing water, the river follows the local steepest descent, but clearly there are paths that descend much more rapidly without taking all the meanders that the river does. Nevertheless, the path of steepest descent is, on the average, a good way to go to the local minimum.

Since nearby paths all go to the same minimum, it is not necessary to follow the path of steepest descent closely—if we do not wander too far from it on any one step, we will still end up at the same place. Thus we have a reasonable amount of robustness built into the method of steepest descent.

The method of steepest descent has another weakness, however. Near a minimum the negative gradient points only weakly to the minimum (because of rounding errors in the computation, if for no other reason). But this should not be exaggerated—if the minimum is broad and shallow, then almost any point nearby will give almost as good a response for the function being examined.

43.6 FOLLOWING THE GRADIENT

The negative gradient gives the differential direction of the path of steepest descent—thus the problem of following the path resembles that of integrating a system of n ordinary differential equations. In principle all we need to do is

follow the local negative gradient. The crudest method would be compute the vector whose components are

$$x_j^{(i+1)} = x_j^{(i)} - h \frac{\partial f^{(i)}}{\partial x_j} \quad j = 1, 2, \dots, n$$

where we have used superscripts to indicate the step number and subscripts to indicate the variables, and h is the step size. The method may seem crude, but remember that we do not have to be accurate in our individual steps; all the local paths will end at the same place *independent* of the local errors (provided the errors are not too big).

What step size shall we use? One way is to use the quantity which measures the size of each step

$$\sum_{j=1}^n (x_j^{(i+1)} - x_j^{(i)})^2$$

and if this is too small, then increase the step size. Near the minimum we cannot expect to gain much, even for big steps.

The fault of this method, and all others that use frequent reevaluations of the gradient, is simply that it usually costs a great deal of computing time to evaluate the n components of the gradient. If you insist on using this crude approach that reevaluates the gradient each step, then probably a modified Euler method (Sec. 22.3) would be better.

Rather than following the local gradient each step, it is much more efficient to find a local gradient and then follow that direction down until you meet a minimum on that path (on the surface). At this point you find a new local negative gradient and follow that direction to a local minimum on that line. It is comparatively easy to see that the successive directions you take will be perpendicular to each other. See Fig. 43.6.1.

In more detail, we find the direction of the negative gradient and take a step size h in that direction. We expect to find a smaller value of the objective func-

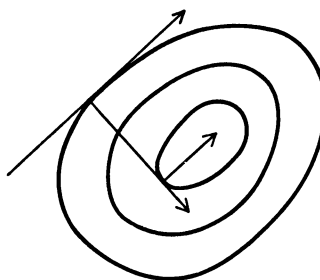


FIGURE 43.6.1

tion at that point, and if we do, then we keep on taking steps until we find an increase in the successive steps. We could then use something like the bisection method to try to find the lowest point on the line. If the first step gives us a value too high, then we have to resort to the bisection method immediately.

Alternately, once we find three points with the middle one lower than the other two, we could use quadratic interpolation and find the minimum of the quadratic as the next choice, rather than use the bisection method (which is guaranteed to be slow, but reliable). This speeds up the search, so that the first phase is to find three points with the middle one lower than the other two. If we find after a number of initial steps that we have not found the three points, then we are tempted to increase our search step size, say double it, and if still we are taking too many steps, then we double again. The process is much like that for doubling the step size for differential equations. Similarly halving can be used if we overshoot the minimum on the first step.

43.7 ESTIMATING THE GRADIENT

In many problems it is totally impractical to find expressions for the local gradient, let alone evaluate them. Instead it is necessary to examine the surface of the objective function and from it to guess at the gradient. To do this we simply make small changes in the coordinate we are interested in, and from the difference of the two values on the surface we estimate the derivative in that direction. For n independent variables we will have to evaluate the surface at n nearby points to get the estimates of the n partial derivatives. Furthermore, the choice of nearby points means that the estimates will be mainly roundoff, while placing the points far apart means that we will not be getting the local derivative. Either way you can lose!

The method and its problems are so obvious that the details of what can happen to you and how to cope with them need not be discussed here. For example, if the point where you are is a saddle point, then you can be badly fooled. Consider the surface

$$z = xy \quad \text{at } x = 0, y = 0$$

We have

$$\frac{\partial z}{\partial x} = 0 \quad \text{and} \quad (\Delta z)_x = 0$$

$$\frac{\partial z}{\partial y} = 0 \quad \text{and} \quad (\Delta z)_y = 0$$

so that you would not have a reliable indication of the (local) surface and where to go next.

43.8 SOME PRACTICAL OBSERVATIONS

The valleys that seem to cause the most trouble when hunting for a local minimum are long, thin, curving ones that occur much more often than seems reasonable at first glance. As Fig. 43.8.1 shows, for such valleys the search path is not efficient, and it runs back and forth. Thus for such valleys the direct search can be expensive.

Our trouble is that we do not have an invariant algorithm (Sec. 4.8) for our search. To see this most clearly, consider an elliptical valley where most of the time the gradient does not point to the minimum. If we could find the transformation that opened up the valley into a circle, then we would get to the minimum

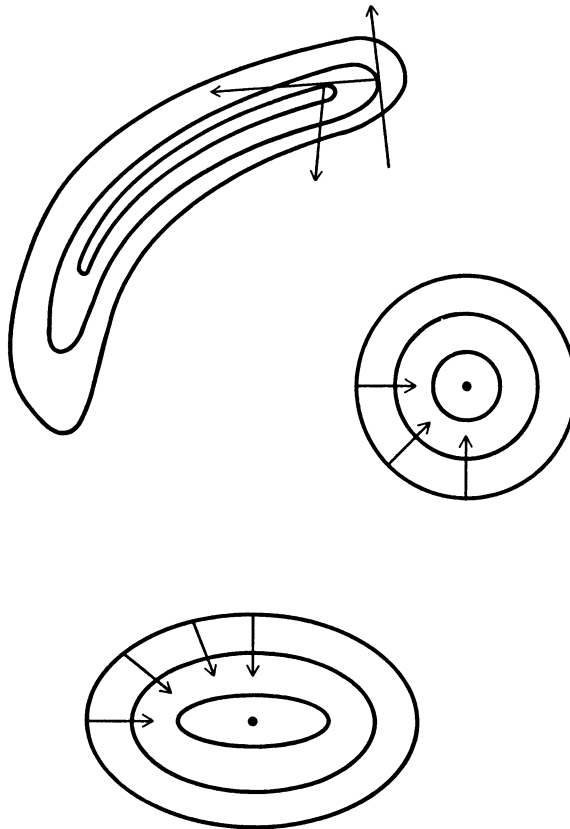


FIGURE 43.8.1

in the search along the first gradient line rather than having to shift from gradient line to gradient line. We shall look briefly at this in the next section on the variable metric methods. These methods attempt to find the shape of the valley in the large by keeping information about the shape, using where the path has been rather than depending completely on the local topography (as does the method of steepest descent).

A valley (ridge) that is not parallel to a coordinate axis and is not straight is very common and difficult to follow. The so-called "ridge-following methods" seem to work sometimes, but the better they work on a given type of ridge, the worse they are apt to be in finding a ridge in the first place. They should be used only with caution.

43.9 THE FLETCHER-POWELL METHOD

As we observed, Newton's method may be very good when the zero is successfully isolated (though there is no reasonable test to know when this happens), but it is apt to be slow when far from a zero. Hence methods which attempt to fit the surface with a quadratic and then find the minimum of the quadratic work fine when near the extreme value, but they are not very good when far away. Thus the variable metric methods tend to be used near the end of the search rather than at the beginning.

Newton's method in one variable uses the function and the first derivative to get the next value. Since the minimum is a zero of the derivative, a quadratic search method will involve the derivative and the second derivative which is the Hessian. In fact it is the reciprocal of the matrix of second partial derivatives that is needed. If we had this inverse, and if the surface were exactly a quadratic, then we would be able to find the minimum in one step. In a sense it is an attempt to produce an invariant algorithm that will take the correspondingly scaled steps for any of the multiplicative transformations that leave floating-point numbers with the same accuracy.

The Fletcher-Powell method¹ seems to be the most popular of the quadratic convergence methods. In order to follow the process, we introduce the notation

$$(x, y) = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n = \text{scalar}$$

and

$$\langle x, y \rangle = (x_i y_j) = \text{matrix}$$

where x and y are vectors having n components.

¹ R. Fletcher, and M. J. D. Powell, A Rapidly Convergent Descent Method for Minimization, *Computer Journal*, vol. 6, pp. 163-168, 1963.

We suppose that the surface is a quadratic in n variables

$$\begin{aligned} f &= f_0 + \sum_{i=1}^n a_i x_i + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n G_{i,j} x_i x_j \\ &= f_0 + (a, x) + \frac{1}{2} (x, Gx) \end{aligned}$$

where, of course, G is the matrix (G_{ij}) and is taken to be symmetric. We want G^{-1} and start with the positive definite¹ matrix

$$H^{(0)} = I$$

where I is the identity matrix.

Let the current point be $x^{(i)}$, with gradient $g^{(i)}$ and current matrix $H^{(i)}$. Set

$$S^{(i)} = -H^{(i)}g^{(i)}$$

and search in this direction until you find a minimum which defines α by

$$f(x^{(i)} + \alpha S^{(i)}) = \text{minimum}$$

Set

$$\begin{aligned} \sigma^{(i)} &= \alpha^{(i)} S^{(i)} \\ x^{(i+1)} &= x^{(i)} + \sigma^{(i)} \end{aligned}$$

Evaluate

$$f(x^{(i+1)}) \quad \text{and} \quad g^{(i+1)}$$

Set

$$\begin{aligned} y^{(i)} &= g^{(i+1)} - g^{(i)} \\ H^{(i+1)} &= H^{(i)} + A^{(i)} + B^{(i)} \end{aligned}$$

where

$$\begin{aligned} A^{(i)} &= \frac{\langle \sigma^{(i)}, \sigma^{(i)} \rangle}{(\sigma^{(i)}, y^{(i)})} \\ B^{(i)} &= \frac{\langle -H^{(i)}y^{(i)}, y^{(i)}H^{(i)} \rangle}{(y^{(i)}, H^{(i)}y^{(i)})} \end{aligned}$$

It can be shown that the matrix $H^{(i)}$ remains positive definite and that the convergence is quadratic. We shall not go into the details since they involve a good deal of linear algebra and are available in the original paper.

In practice it is well to restart the matrix H after n or possibly $2n$ steps to get rid of the old estimates of the surface that were far from the minimum.

¹ Positive definite matrix means that for all $(x, x) \neq 0$ then $(x, Hx) > 0$.

43.10 OPTIMIZATION SUBJECT TO LINEAR CONSTRAINTS

Up to now the problems in this chapter have been problems of unconstrained optimization. This has meant that in our search for a minimum of $f(x_1, \dots, x_n)$, any value of the variables $x_1, \dots, x_n$ was a permissible one. There are many practical cases, however, in which physical or mathematical reasoning forces us to restrict our search to values of the variables which satisfy certain conditions.

These conditions, or constraints as we will call them from now on, can take on many forms. In this section and the next we shall examine some of the forms in which constraints appear and how they affect the solution, and we shall suggest methods for handling them.

Some constraints are easier to handle than others. Consider, for example, nonnegative constraints: that is, each of the variables $x_1, \dots, x_n$ is restricted to nonnegative values. Under these conditions, the statement of the optimization problem would read: Minimize the function

$$f(x_1, \dots, x_n)$$

subject to the constraints

$$x_i \geq 0 \quad i = 1, \dots, n$$

The easiest way to solve this problem is through a transformation of variables. If we let

$$x_i = y_i^2 \quad i = 1, \dots, n$$

then we may minimize f as an unconstrained function of the $y_1, \dots, y_n$ and be certain that the nonnegative constraints will be satisfied at the optimum. An alternative way is to apply the gradient or the quasi-Newton technique of Sec. 43.9 with the following modification: Whenever a variable, say x_k , becomes zero and at the same time the k th component of the gradient is positive, then stop updating x_k until $\partial f / \partial x_k$ changes sign. This is another way in which we can be sure that the search will be restricted to nonnegative values.

The above tricks apply equally well when the variables are restricted to lying between lower and upper bounds, that is, when the constraints are of the form

$$\begin{aligned} x_i - l_i &\geq 0 \\ u_i - x_i &\geq 0 \end{aligned} \quad i = 1, \dots, n$$

where the l_i and u_i are given and, of course, $u_i > l_i$. Here, we may again use a transformation on the variables. We may, for example, let

$$x_i = (u_i - l_i) \sin^2 y_i + l_i$$

and solve the problem as unconstrained in terms of the y 's. Other transformations are possible; however, one has to be careful about introducing new local minima. This is because the minimization of $f(y_1, \dots, y_n)$ will stop whenever $\partial f / \partial y_i = 0 (i = 1, \dots, n)$. But by the chain rule,

$$\frac{\partial f}{\partial y_i} = \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial y_i}$$

Therefore, any transformation on the x 's which causes $\partial x_i / \partial y_i$ to be zero at a point which meets this constraint and is not on the boundary will introduce new minima.

The constraint types discussed so far are linear and are special cases of the general linear constraints of the form

$$\sum_{i=1}^n a_{ij} x_i - b_j \geq 0 \quad j = 1, \dots, m$$

Linear constraints are much easier to handle than the nonlinear ones, which we discuss in the next section. The main reason for this is that the boundary of the feasible region is composed of straight lines in two dimensions, planes in three dimensions, and hyperplanes in higher dimensions. It is, consequently, easy to travel along a boundary during the search for the optimum. Several methods, developed especially for linearly constrained problems, take full advantage of this fact. Best known among these is the so-called gradient-projection method. This method has been designed primarily for linearly constrained problems with a nonlinear objective function; it is essentially an extension of the steepest-descent method to linear constraints. This presentation is too lengthy to be included in this book and the reader is referred to the literature.¹

A very important class of optimization problems is the one in which both the objective function and all the constraints are linear. Problems of this type which minimize

$$f(x) = \sum_{i=1}^n c_i x_i$$

subject to

$$\sum_{i=1}^n a_{ij} x_i - b_j \geq 0 \quad j = 1, \dots, m$$

are called *linear programming* problems, and they are important because of the enormous variety of applications which they have found in practical situations.² A little thought will make it intuitively clear that the solution of a linear program

¹ Read, for example, the section beginning on p. 133 of T. L. Saaty and J. Bram [50].

² See, for example, S. I. Gass [15].

lies on the boundary of the feasible region, usually on a corner, but sometimes along one side of a constraint. This observation provides an efficient technique for solving linear programs. More details can, again, be found in the literature.

43.11 OTHER METHODS

The Lagrange multiplier approach is a classical one and has endured a long time because of its practical and theoretical importance. There are occasions, however, when the solution of the nonlinear equations resulting from this approach is quite difficult. In such cases, other methods are preferred in practice.

One of the best-known alternative techniques for handling equality constraints was originated by R. Courant. In this method we minimize the function

$$F(x_1, \dots, x_n) = f(x_1, \dots, x_n) + r \sum_{i=1}^m [g_i(x_1, \dots, x_n)]^2 \quad r > 0$$

as an unconstrained problem for a sequence of monotonically increasing values of r . As r becomes infinitely large, the sum of the squares of the constraints is forced to zero. In this way, the sequence of the successive minima of F converges toward the solution of the given problem.

We conclude this chapter with a few brief comments about nonlinear inequality constraints of the form

$$g_i(x_1, \dots, x_n) \geq 0 \quad i, \dots, 1, \dots, m$$

First let us examine the necessary conditions for the solution. It turns out that here, as for the equality constraints, necessary conditions can be expressed through a Lagrange function. If a point $x_1, \dots, x_n$ is a minimum of the given objective function $f(x_1, \dots, x_n)$ subject to the above constraints, then it must be true that the negative gradient vector of f can be expressed as a linear combination of the negative gradients of the constraints which are binding. Geometrically this means that if the minimum point is on the boundary of the feasible region, then the negative gradient of f must fall within the cone which is formed by the gradients of the constraints which are binding, Fig. 43.11.1.

A little thought will show that if this were not true, then we should be able to move a little farther along the boundary to obtain a point at which f is smaller.

There are two common philosophies in the various solution approaches suggested for problems with nonlinear inequality constraints: "the boundary-following approaches" and the "penalty-function techniques."

As their name implies, the boundary-following approaches suggest that when a constraint is (or is about to be) violated, then follow the boundary of the

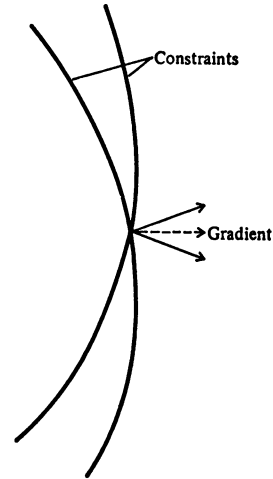


FIGURE 43.11.1

feasible region defined by that constraint until a point which satisfies the necessary conditions for a minimum is found. If the boundary is highly nonlinear, then the convergence of such approaches will be slow. In such cases, we prefer the penalty-function techniques.

The idea of a penalty function was introduced earlier when we added the sum of the squares of equality constraints to f in order to “penalize” the minimization of the resulting function whenever a constraint was violated. A similar technique can be used for inequality constraints. Consider, for example, the functions

$$F(x_1, \dots, x_n) = f(x_1, \dots, x_n) + r \sum_{i=1}^m \frac{1}{g_i(x_1, \dots, x_n)} \quad \begin{array}{l} r > 0 \\ r \rightarrow 0 \end{array}$$

or

$$F(x_1, \dots, x_n) = f(x_1, \dots, x_n) - r \sum_{i=1}^m \ln[g_i(x_1, \dots, x_n)] \quad r \rightarrow 0$$

or

$$F(x_1, \dots, x_n) = f(x_1, \dots, x_n) + r \sum_{i=1}^m [\tilde{g}_i(x_1, \dots, x_n)]^2 \quad r \rightarrow \infty$$

where $\tilde{g}_i = 0$ if $g_i \geq 0$, and $\tilde{g}_i = g_i$ if $g_i < 0$.

If we minimize these functions sequentially for a series of positive values for r (monotonically decreasing for the first two functions and increasing for the third one), then we approach a solution to the constrained problem.

44.1 INTRODUCTION

Linear independence plays a central role in much of mathematics. In mathematics it is a yes/no thing, and there are no measures of the degree of linear independence.

A little thought shows that linear independence is also very important in numerical methods, but as is so often the case, there are probably a number of different ways to measure the degree of linear independence. Unfortunately none of these possible ways has been developed to any extent, let alone the whole organized into a reasonable classification. Thus the best we can do in this chapter is to go through the material presented so far and indicate where various possible ideas might be used, if only we could develop them adequately!

The fact that the material in this chapter, like much of the more recent chapters, is vague and not of the theorem-proof form is not an indication of unimportance, but rather of being near the frontier of the field. In time it is probable that the material will become organized as more definite bodies of knowledge. But the material being at present vague, it is conventional to skip

over it in the standard mathematically inclined texts, and therefore references are sometimes hard to give.

As simple examples of the basic role of linear independence in mathematics, consider the following definitions.

- 1 A number is *transcendental* if the powers of it are linearly independent over the integers (or fractions), that is, if

$$\sum_{k=0}^N a_k x^k = 0$$

implies all the $a_k = 0$; otherwise it is algebraic.

- 2 A function $y = y(x)$ is transcendental if the powers are linearly independent over the ring of polynomials, that is,

$$\sum_{k=0}^N P_k(x) y^k(x) \equiv 0$$

implies all the polynomials $P_k(x) \equiv 0$.

44.2 LINEAR EQUATIONS

In Chap. 7 we discussed the solution of a system of linear algebraic equations, and we went to considerable trouble to point out some of the faults of the widely accepted methods of solution. The methods tend to avoid mentioning the linear dependence (except for rank) of the equations. An ill-conditioned system is recognized as being somewhat dependent, but the direct approach to creating a measure of independence is avoided (condition numbers are hardly a measure of it).

We would like to have a measure of independence for a set of equations. The values could run from 1 to ∞ (dependent) much as entropy does, or they could run from 1 to 0 as probability does. One intuitively feels that dependence is like entropy: every operation that is done on the system can only increase the dependence. This is consistent with our basic feelings on the matter—it should not be possible by any mathematical operations to increase the independence of the system, but it should be possible to decrease it. Reformulation of the problem should be the only way to increase the independence.

The use of the angle between the lines or planes is often thought to be relevant, but Prob. 7.9.2 discusses the case for just two equations. The solution shows that a floating-point transformation by a power of 2 in one variable changes the angle but clearly leaves the equations as independent as before. For two lines with slopes of opposite sign there is always a transformation that makes the lines

perpendicular; but when the slopes are the same sign, there is a maximum angle that can be achieved. Whether this is relevant or not and how it works in n dimensions are not clear to the author.

If we keep the vague general idea of linear dependence in our minds and ask how to make it as large as we can in one step of elimination of simultaneous linear equations, then we decide that using *the same equation* to eliminate from all the others would do the best job—but this is pivoting, the most widely used method! Thus one can only wonder about the wisdom of pivoting—yes, at the current step it seems to have some nice roundoff properties, but it also seems to be designed to increase the linear dependence (assuming we do have an intuitive idea of its meaning), and hence at a later stage we may come to regret it. The truth of this observation is shown in the example of Sec. 7.8 of solving the 3×3 symmetric system that is apparently well scaled, and for which Gaussian elimination produces dependence while another method keeps the independence and permits the accurate solution of the system.

How could we proceed other than by pivoting? We could try eliminating by pairs, thus not favoring one particular equation which runs the risk that later everywhere in the derived system we will “see” the same pivot equation staring at us. This method still tends to favor the two end equations which will enter into only one equation each, while all the other equations will enter into two derived equations. This could be coped with by imagining the equations on a cylinder, and eliminating by pairs to get n equations in one less variable at each stage. At the end we would have n equations in one unknown which could be solved by least squares (the mean) for the most probable value. Putting this in the earlier system we again have n equations in the next unknown, and so we go “least squaring” ourselves up the back solution equations one at a time. Almost nothing is known about such a method except the obvious fact that it will take about twice as long in machine time to do as the conventional Gaussian elimination method and a good deal of storage as well.

Frankly the author does not know how to solve linear equations ideally, but feels strongly that any method of analysis that does not come to terms with the linear independence of the system, and the loss per step, is likely to be superficial. Linear independence is too basic in such a problem to be ignored. The difficulty, of course, is to find an adequate, relevant measure of dependence of a system of equations.

One final word of suggestion for doing what the author does not know how to do. When framing definitions of linear independence (dependence), keep in mind that we use floating-point arithmetic of a fixed precision (number of digits), and are not dealing in the usual mathematical fixed-point arithmetic. If this is kept in mind, it is soon seen that several of the arguments usually given about the

size of the roundoff do not apply. Thus the idea of taking the largest coefficient as a pivot in order to reduce the roundoff does not hold. For suppose that with only two equations we use the smaller to eliminate from the larger. The result is, of course, an equation with larger coefficients (hence larger fixed-point roundoff) but the same floating-point roundoff as the other way round. It is the way the equations are combined, not the coefficients, that matters. Of course, in the usual elimination process the final equation for each x_i is the same linear combination of the original equations *regardless* of the process along the way, but the roundoff effects of the recombinations of the same equation can be considerably different.

44.3 SAMPLING AND LINEAR INDEPENDENCE

For each family of functions, polynomials (the Vandermonde determinant in Sec. 14.4), the Fourier series (orthogonality in Sec. 31.5), and the exponentials (Sec. 39.2), we had to prove the very important result that the mathematical linear independence of the functions over any continuous interval could be translated into linear independence of n functions over the set of n distinct samples. *This is the basis for the validity of the numerical approach.*

If we ask, as we do in this chapter, for the *amount* of linear independence, then we have to describe a measure of it. For polynomials it is natural to ask where we can put the sample points x_i so that the crucial Vandermonde determinant has the largest possible value for its square. The farther apart the points, of course, the larger the value, and it is necessary to normalize the question to get a reasonable result. It is natural, again, to ask that the sample points x_i be confined to some fixed interval, say $-1 \leq x \leq 1$.

Given this problem, we can easily see that for two points the ends of the interval are best, for three the ends and the middle. For four points, using symmetry, we want to maximize the product

$$\begin{aligned} V_4 &= \sum_{i \geq j=1}^4 (x_i - x_j)^2 & x_1 &= -1 \\ &= [4x_2(1 - x_2^2)^2] & x_4 &= +1 \\ & & x_2 &= -x_3 \end{aligned}$$

Equating the derivative to zero and finding the maximum, we get

$$\begin{aligned} 5x_2^2 - 1 &= 0 \\ x_2 &= 1/\sqrt{5} = -x_3 \end{aligned}$$

The next case is clearly

$$\begin{aligned}x_1 &= -1 \\x_2 &= -x_4 \\x_3 &= 0 \\x_5 &= 1 \\V_5 &= 4x_2^3(1 - x_2^2)^2\end{aligned}$$

which leads to

$$7x_2^2 - 3 = 0 \quad \text{or} \quad x_2 = \pm\sqrt{3/7}$$

It is futile to go on case by case, so we ask how could the samples be found in general. Clearly they can come from some family of polynomials. We have, so far

Vandermonde	Polynomial [less $(1 - x^2)$]
$V_2 = (1 - x^2)$	$p_0 = 1$
$V_3 = x(1 - x^2)$	$p_1 = x$
$V_4 = 4x(1 - x^2)^2$	$p_2 = 5x^2 - 1$
$V_5 = 4x^3(1 - x^2)^2$	$p_3 = x(7x^2 - 3)$

Where we can expect to get the polynomials from? Probably from some orthogonal system. The first try, using weight factor $1 - x^2$ and, of course, range $-1 \leq x \leq 1$, is

$$\int_{-1}^1 (1 - x^2)p_i(x)p_j(x) dx = 0 \quad i \neq j$$

and shows that they are indeed orthogonal (we need only do p_0p_2 and p_1p_3). Thus we have ultraspherical polynomials (Sec. 26.2 with $\alpha = 1$). A check of Szego¹ shows that we are correct in our heuristic approach; the zeros of the ultraspherical polynomials are indeed the sample points that maximize the Vandermonde, and hence in this sense give the maximum linear independence. Of course, the Gaussian quadrature points give another version of where to pick the sample points when integrating.

PROBLEMS

44.3.1 Show that the next polynomial is $p_4 = 21x^4 - 14x^2 + 1$ and check the orthogonality of it with respect to p_0, p_1, p_2, p_3 .

¹ G. Szego, *Orthogonal Polynomials*, *Am. Math. Soc.*, vol. 23, Colloquium Publications, 1959, p. 139.

44.4 POWERS OF x

The size of the Vandermonde determinant gives one measure of the linear independence of the powers of x over a set of points. There are other measures of independence. Consider again the interval $-1 \leq x \leq 1$ and a linear combination of the powers of x , where to normalize the problem we take the first coefficient equal to one

$$p_n(x) = x^n + a_{n-1}x^{n-1} + \cdots + 1$$

The choice of the coefficients of the n th Chebyshev polynomial (times 2^{n-1}) gives the inequality

$$|p_n(x)| = \left| \frac{T_n}{2^{n-1}} \right| \leq \frac{1}{2^{n-1}} \quad -1 \leq x \leq 1$$

and shows that the powers of x are practically dependent for large n . This agrees with our intuitive feelings that the even powers of x look too much like each other to be very independent, and similarly for the odd powers (notice that each Chebyshev polynomial used only one set of these).

Thus the choice of the powers of x (for the interval $-1 \leq x \leq 1$) as the basis for the representation can lead to trouble because they are not very independent. The fact that the powers are easily computed on a computer does not mean that they should be used indiscriminately. Rather the orthogonal polynomials, discussed again in the next section, provide a better basis, and using the three-term recurrence relation (Sec. 26.6) they are not difficult to compute for moderate n .

44.5 ORTHOGONAL POLYNOMIALS AND LEAST SQUARES

Orthogonal polynomials and least squares provide an alternate approach to the problem of measuring the linear independence of a set of functions. Section 26.3 shows that orthogonal functions are linearly independent.

If we rate the linear independence of the powers of x by the least-squares measure

$$\left[\int_{-1}^1 p_n^2(x) dx \right]^{1/2}$$

where

$$p_n(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$$

we find that corresponding to the Chebyshev polynomial of the previous section we get the Legendre polynomials $P_n(x)$ (Sec. 26.7). The corresponding proof is found by writing

$$p_n(x) = c_n P_n(x) + c_{n-1} P_{n-1}(x) + \cdots + P_0(x)$$

where we know that $c_n = n!/[1 \cdot 3 \cdot 5 \cdots (2n-1)]$. The integral, due to the orthogonality, is

$$\int_{-1}^1 p_n^2(x) dx = \sum_{k=0}^n c_k^2 \cdot \frac{2}{2k+1}$$

and this is clearly minimized by taking $c_0 = c_1 = \cdots = c_{n-1} = 0$. Thus the least-squares measure is

$$\begin{aligned} \sqrt{\frac{2}{2n+1}} c_n &= \frac{n! n! 2^n}{(2n)!} \sqrt{\frac{2}{2n+1}} \\ &\simeq \frac{(n^n e^{-n})^2 2\pi n \cdot 2^n}{(2n)^{2n} e^{-2n} \sqrt{2\pi \cdot 2n} \sqrt{n+1/2}} \\ &\simeq \frac{\sqrt{\pi}}{2^n} \end{aligned}$$

which is about the same as the Chebyshev measure.

On the other hand consider the linear independence of the orthogonal functions. For purposes of comparison we shall suppose them to be normalized, in the interval $-1 \leq x \leq 1$, that is,

$$\int_{-1}^1 f_i(x) f_j(x) dx = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

Consider now the linear combination

$$c_0 f_0(x) + c_1 f_1(x) + \cdots + c_n f_n(x)$$

where all the c_i are not zero. It is natural to normalize the linear combination by requiring that

$$\sum_{i=0}^n c_i^2 = 1$$

Thus our measure now becomes

$$\int_{-1}^1 [c_0 f_0(x) + c_1 f_1(x) + \cdots + c_n f_n(x)]^2 dx = \sum_{k=0}^n c_k^2 = 1$$

for *all* admissible choices of the c_i .

In this framework of ideas we see that the orthogonal functions, including orthogonal polynomials, generally have a nice, uniform amount of linear independence and why they so often form a basis of representation preferable to other

mathematically equivalent bases. (See Hilbert matrix arguments in Sec. 25.6.) The equations for the least-squares fit are a nice diagonal set of orthogonal functions and can be very ill-conditioned for the basis $1, x, x^2, \dots, x^n$.

In still another light, each of the orthogonal polynomials has all its zeros real, distinct, in the interval, and interlacing those of the adjacent polynomials, while the even (odd) powers of x tend to resemble each other more and more as the degree rises. This again gives an indication that for many purposes the orthogonal polynomials form a reasonable basis for representation, and the original powers of x do not. The problem of the basis for representation of our problem is rarely discussed in detail, though we did observe, in Sec. 13.7, the difficulty of having the wrong function get into the basis for the representation and (in Sec. 7.8) the lack of independence in the $\sinh$ and $\cosh$ representation of functions for large arguments, while the mathematically equivalent e^x and e^{-x} are fine. Near the origin, however, the $\sinh x$ and $\cosh x$ are more different than the exponentials, so that there is no magic answer for all problems.

The quasi-orthogonal polynomials introduced in Sec. 27.3, provide an example of how to choose a basis for representing functions without paying the high price of trying to make them exactly orthogonal (using perhaps the impractical Schmidt process, Sec. 26.3).

44.6 WHAT SAMPLES?

Two of the fundamental problems in numerical analysis are what samples and what functions, and these are related. We have discussed the effect of sample points on polynomials, and the phenomenon of aliasing (Sec. 31.2) shows the effect on sinusoids. The study of integration formulas using one or more derivatives (Sec. 17.8) shows that a sample of another derivative at a point where the function and the lower derivatives are already known is almost as good as another sample of the function at a new point.

There are versions of the sampling theorem (Sec. 34.6) that use function values and derivatives, and again one sees that the derivative at an old point is about as good as a new function value.

44.7 WHICH BASIS OF FUNCTIONS?

The problem of the basis for representation of the solution is central, yet almost all mathematics is content to affirm that the linear independence is all that matters. In the solution of difference (Chap. 13) and differential equations (Chaps. 23 and

24) this can make a significant difference. In the neighborhood of singularities (Chap. 42) and even in the representation of functions (Chap. 3), for example

$$f(x) = \sqrt{x+1} - \sqrt{x} \equiv \frac{1}{\sqrt{x+1} + \sqrt{x}}$$

the form used to represent the function is vital to good computing.

The literature constantly shows that the proper, “natural” set of functions gives spectacular improvements in accuracy and decreases in cost. It is a matter of studying the problem on hand to see which representation is preferable, and many times this depends mainly on the linear independence of the chosen set.

45

EIGENVALUES AND EIGENVECTORS OF HERMITIAN MATRICES¹

45.1 WHAT ARE EIGENVALUES AND EIGENVECTORS?

A vector x in Euclidean n -dimensional space has n components x_i , ($i = 1, \dots, n$). In calculus we studied (and in Sec. 26.1 we referred to) the product of two n -dimensional vectors x and y , denoted by

$$(x, y) = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n = xy$$

If $(x, y) = 0$, we say (again see Sec. 26.1) that x and y are *orthogonal*. A convenient measure of the “length” of a vector x is the Euclidean norm

$$|x| = \text{norm of } x = (x, x)^{1/2} = (\sum x_i^2)^{1/2}$$

When the norm of a vector is 1, it is called a *unit vector*.

A square matrix $A = (a_{i,j})$ multiplied by a column vector x produces a column vector y :

$$Ax = y$$

¹ This chapter owes its good points to M. P. Epstein as well as to conversations with P. A. Businger and notes from C. B. Moler.

Thus, matrix multiplication is a linear transformation of the points (vectors) in Euclidean n -dimensional space into points of the same space. The null vector (every component is 0) is usually written as 0 and goes into itself. It is unusual for any other point (vector) to go into itself *except* when $A = I$, the identity matrix with all 1s on the main diagonal and 0s elsewhere, in which case *every* vector goes into itself.

While vectors do not go into themselves, most of the time a square matrix of order n will have n different directions which go into themselves; that is, there are (usually) n different values of a scalar λ for which the following system of equations can be solved:

$$Ax = \lambda x$$

each, of course, giving rise to a different vector x . This is not surprising, since writing the equations in the form

$$(A - \lambda I)x = 0$$

indicates that we can solve the system if and only if the corresponding determinant is zero; namely, if and only if

$$|A - \lambda I| = 0$$

By inspection this determinant is a polynomial of degree n in λ , and in general it will have n distinct zeros, real or complex.

Why should we care about these special values of λ and their corresponding solutions, which, to introduce the jargon, are called, respectively, eigenvalues and eigenvectors? It will turn out that for the cases we are interested in (symmetric and Hermitian matrices) the n eigenvectors are linearly independent and form an orthogonal basis. Thus, an arbitrary vector x can be written in the form

$$x = c_1 x^{(1)} + c_2 x^{(2)} + \cdots + c_n x^{(n)}$$

where the superscript j is used to denote the j th eigenvector. The coefficients c_j are given by

$$c_j = \frac{(x, x^{(j)})}{(x^{(j)}, x^{(j)})} = \frac{(x, x^{(j)})}{|x^{(j)}|^2}$$

When this vector x is multiplied by the matrix A , the individual components of the representation are merely multiplied by their corresponding eigenvalues, that is,

$$Ax = c_1 \lambda_1 x^{(1)} + c_2 \lambda_2 x^{(2)} + \cdots + c_n \lambda_n x^{(n)}$$

Thus, the effect of matrix multiplication by A is particularly easy to follow in the eigenvector representation of an arbitrary vector, since it is a combination of

“stretchings” (or “contractions”) by factors of λ_j along the axes $x^{(j)}$. This is much of the reason that eigenvalues and eigenvectors play an important role in many applications.

45.2 NOTATION AND HERMITIAN MATRICES

The transpose of a matrix

$$A = (a_{i,j})$$

is written

$$A^T = (a_{j,i})$$

that is, the transpose A^T is obtained from A by writing the rows of A as the columns of A^T (or else the columns as rows) in the same order. Thus, if x is a column vector, then x^T is a row vector, and

$$(x, y) = y^T x = \sum_{i=1}^n y_i x_i$$

$$|x| = (x^T, x)^{1/2}$$

The outer product of two vectors, defined by

$$\langle x, y \rangle = x \cdot y^T$$

is an $n \times n$ matrix (of rank 1) in which the element in the i th row and j th column is $x_i y_j$. This is in contrast with the earlier (inner) product of two vectors, which is a scalar.

It is easy to see from the definition of matrix multiplication that

$$(AB)^T = B^T A^T$$

that is, the transpose of a product is the product of the transposes in reverse order. Finally, it is obvious that

$$(A^T)^T = A$$

All the above can be generalized to matrices with complex elements by making appropriate changes. The central idea is the *conjugate transpose* of a matrix of complex elements in place of the earlier transpose. The conjugate transpose is

$$A^H = \bar{A}^T = (\bar{a}_{j,i})$$

A matrix for which $A^H = A$ is called Hermitian in honor of the mathematician Hermite. For such a matrix, if¹

$$a_{i,j} = p_{i,j} + iq_{i,j}$$

¹ Do not confuse the coefficient $i = \sqrt{-1}$ with the subscript i .

then by definition

$$a_{j,i} = p_{i,j} - iq_{i,j}$$

In words, the real part of the Hermitian matrix is symmetric while the imaginary part is “skew-symmetric.”

For complex valued vectors the inner product can be defined as

$$(x, y) = y^H x$$

and again the norm is real since

$$|x| = (x, x)^{1/2} = \text{a real scalar}$$

We also have, as before,

$$(AB)^H = B^H A^H \quad \text{and} \quad (A^H)^H = A$$

Perhaps the most important property of Hermitian matrices is that all their eigenvalues are real. To see this, let x be any eigenvector with eigenvalue λ . Then

$$Ax = \lambda x$$

Since $A^H = A$,

$$A^H x = \lambda x$$

Therefore, taking the complex transpose of both sides,

$$(A^H x)^H = (\lambda x)^H \quad \text{or} \quad x^H A = \bar{\lambda} x^H$$

Multiply this equation on the right by x and the original equation on the left by x^H to get the pair

$$x^H Ax = \lambda x^H x$$

$$x^H Ax = \bar{\lambda} x^H x$$

Thus $\lambda = \bar{\lambda}$ and λ is real.

We next show that for any two distinct eigenvalues λ_1 and λ_2 of a Hermitian matrix the corresponding eigenvectors are orthogonal. We have

$$Ax^{(1)} = \lambda_1 x^{(1)}$$

$$Ax^{(2)} = \lambda_2 x^{(2)}$$

Multiply the first equation on the left by $x^{(2)H}$ and the second on the left by $x^{(1)H}$:

$$x^{(2)H} Ax^{(1)} = \lambda_1 x^{(2)H} x^{(1)}$$

$$x^{(1)H} Ax^{(2)} = \lambda_2 x^{(1)H} x^{(2)}$$

Take the conjugate transpose of the first of these equations (recall that $\bar{\lambda}_1 = \lambda_1$, and $A^H = A$), and subtract from the second equation to get

$$(\lambda_2 - \lambda_1)x^{(1)H}x^{(2)} = 0$$

Since we assumed that $\lambda_2 \neq \lambda_1$, we have

$$(x^{(2)}, x^{(1)}) = x^{(1)H}x^{(2)} = 0$$

and the vectors are orthogonal.

From this it follows that if a Hermitian matrix has all its eigenvalues distinct then the eigenvectors form an orthogonal basis for Euclidean n -dimensional space. (When the eigenvalues are not all distinct it is still possible to find n independent, orthogonal eigenvectors, but we will not prove this.)

45.3 SIMILARITY REDUCTIONS

The formality of going from symmetric to Hermitian matrices is straightforward. Since it is symmetric matrices that occur most often, and the necessity of using complex conjugate obscures what is going on in the general case, we shall confine the remainder of the chapter to the case of real symmetric matrices.

Suppose that a symmetric matrix A has n eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (distinct or not) and let D be the diagonal matrix

$$D = \begin{pmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{pmatrix}$$

Let X be the $n \times n$ matrix whose columns are the orthogonal eigenvectors $x^{(1)}, x^{(2)}, \dots, x^{(n)}$. Then the eigenequations

$$Ax^{(i)} = \lambda_i x^{(i)} \quad i = 1, 2, \dots, n$$

can be written in the more compact form of matrices

$$AX = XD$$

where D is on the right because it multiplies the columns. The linear independence of the eigenvectors means that the inverse X^{-1} exists and we can multiply on the left to get

$$X^{-1}AX = D$$

Definition: If for any nonsingular matrix S we have

$$B = S^{-1}AS$$

then we say that B is *similar* to A .

Clearly we can suitably multiply on the left and right to get

$$A = SBS^{-1}$$

and A is also similar to B . It is easy to show that similarity is a formal equivalence relation.

The importance of similarity lies in the fact that similar matrices have the same eigenvalues, though the eigenvectors are in general different. To prove this, suppose that

$$Ax = \lambda x \quad \text{and} \quad B = S^{-1}AS$$

Multiply the first equation on the left by S^{-1} , and insert $SS^{-1} = I$ between A and x . We get as a result

$$(S^{-1}AS)(S^{-1}x) = \lambda(S^{-1}x)$$

which is the same as

$$B(S^{-1}x) = \lambda(S^{-1}x)$$

Thus λ is an eigenvalue of B with associated eigenvector $(S^{-1}x)$.

The eigenvector matrix X appearing above has an additional interesting property. We have shown that there exist mutually orthogonal eigenvectors in the case of distinct eigenvalues (and asserted that even when they are not distinct there is an orthogonal set of eigenvectors), so that

$$(x^{(i)}, x^{(j)}) = 0 \quad \text{for} \quad i \neq j$$

If in addition the set of eigenvectors is orthonormal (Sec. 26.1), then each eigenvector is a unit vector. We can always make them orthonormal, since any non-zero multiple of an eigenvector is still an eigenvector. We then have

$$(x^{(i)}, x^{(i)}) = 1 \quad \text{for all } i$$

These two sets of inner-product equations are summarized by the matrix equation

$$XX^T = X^T X = I$$

so that we have the important result

$$X^{-1} = X^T$$

Thus, if we normalize our eigenvectors to be unit vectors, we can simplify the similarity relationship between the matrices A and D to

$$X^T A X = D$$

It is natural to call matrices with this property, $X^{-1} = X^T$, *orthogonal* matrices. We will discuss them further in the next section. (In the complex case, a matrix for which $X^H = X^{-1}$ is called *unitary*.)

PROBLEMS

45.3.1 Show that similarity is an equivalence relation (reflexive, symmetric, and transitive).

45.3.2 Carry out the details of the section for the complex case.

45.4 ORTHOGONAL TRANSFORMATIONS

We just defined a matrix Q for which

$$Q^T = Q^{-1}$$

to be orthogonal. To get an idea of why the word orthogonal is used, consider the classic orthogonal transformation

$$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

which represents a plane rotation through an angle θ . This matrix transforms any pair of orthogonal axes into another pair of orthogonal axes, from which comes the name. This matrix also has the property that its transpose is its inverse.

The $n \times n$ matrix

$$\begin{pmatrix} 1 & 0 & \dots & \dots & \dots & \dots & \dots \\ 0 & 1 & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \cos \theta & \dots & \dots & \dots & \sin \theta \\ \dots & \dots & \dots & 1 & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & 1 & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & 1 & \dots \\ \dots & \dots & -\sin \theta & \dots & \dots & \dots & \cos \theta \end{pmatrix}$$

also performs a plane rotation, but in Euclidean n -dimensional space. The matrix

$$\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$$

is a rotation *plus* a reflection (the determinant is -1). This is also an orthogonal transformation and can likewise be done in n dimensions.

These are the simplest orthogonal transformations since they involve only two coordinates. There are many more orthogonal transformations that are more complicated to write out in detail, and we will look at one particular class of them in the next section.

Orthogonal transformations have the important property that they do not “stretch” the space, that is, if Q is orthogonal, then

$$|Qx| = |x|$$

To prove this, we write the easily verified equation

$$(Ax, y) = (x, A^T y)$$

Replace y by Ay to get

$$(Ax, Ay) = (x, A^T Ay) = (x, y)$$

Now set $y = x$, and we have

$$(Ax, Ax) = (x, x)$$

as required. Thus orthogonal transformations represent only rotations plus reflections but no stretching. Because of this property they seem intuitively more likely to preserve the amount of linear independence of a system fairly well and to hold down roundoff effects, though we shall not try to prove these remarks here.

45.5 HOUSEHOLDER TRANSFORMATIONS

With this background of orthogonal transformations we now turn to the Householder transformation matrices. Let u be a real nonzero vector and let

$$b = \frac{u^T u}{2} = \frac{|u|^2}{2}$$

Then using the outer product $\langle uu^T \rangle = uu^T$, we define a Householder matrix.

$$P = I - \frac{uu^T}{b}$$

For practical computation the above notation is convenient, but for purposes of theory it is simpler to define

$$v = \frac{u}{|u|}$$

so that v is a unit vector and the Householder matrix is

$$P = I - 2vv^T$$

It follows immediately that

$$P^T = I^T - 2(vv^T)^T = I - 2vv^T = P$$

so that P is symmetric. Furthermore,

$$\begin{aligned} P^2 &= (I - 2vv^T)(I - 2vv^T) \\ &= I - 4vv^T + 4vv^Tvv^T \end{aligned}$$

But $v^T v = 1$ so the final terms cancel and

$$P^2 = I \quad \text{or} \quad P^{-1} = P$$

Since P is symmetric, this is the same as

$$P^{-1} = P^T$$

and the Householder matrix P is orthogonal.

These properties are important because for a symmetric matrix A , $P^{-1}AP = PAP$ and this is similar to A . Also this is the same as P^TAP preserving symmetry.

We now examine the effect of multiplying a matrix A by a Householder matrix P . Consider

$$PA = (I - 2vv^T)A = A - 2vv^TA$$

If the components of v are $v_1, v_2, \dots, v_n$ and the rows of A are denoted by $R_1, R_2, \dots, R_n$, then v^TA is the row vector

$$v_1R_1 + v_2R_2 + \cdots + v_nR_n$$

Then $2vv^TA$ is the matrix whose k th row is just

$$2v_k(v_1R_1 + \cdots + v_nR_n)$$

and $PA = A - 2vv^TA$ has the k th row equal to

$$R_k - 2v_k(v_1R_1 + \cdots + v_nR_n)$$

In words, PA is obtained from A by subtracting from each row of A a linear combination of the rows of A . In particular, if $v_k = 0$ then the k th row of PA is the same as the k th row of A .

Similar remarks hold for AP using the columns of A instead of the rows

45.6 TRIDIAGONALIZATION

We will show how to use Householder transformations to reduce a given symmetric matrix to a tridiagonal matrix

$$\begin{pmatrix} \times & \times & 0 & 0 & \cdots & \cdots & 0 \\ \times & \times & \times & 0 & \cdots & \cdots & 0 \\ 0 & \times & \times & \times & \cdots & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & 0 & \times & \times & \times \\ 0 & 0 & 0 & 0 & 0 & \times & \times \end{pmatrix}$$

with the same eigenvalues as A .

To show what Householder transformations can do, we take an arbitrary column vector x with norm σ , and find a Householder matrix P so that Px has specified components the same as those in x and certain other specified components equal to zero. (Of course analogous statements hold for x and xP when x is a row vector.) For purposes of illustrating the flexibility of the method we will suppose that x and P are six-dimensional, in particular,

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix} \quad \text{and} \quad Px = y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{pmatrix}$$

and that we want $y_3 = 0 = y_5$. We cannot accomplish this and have all the other y_i equal to the corresponding x_i (unless we started with $x_3 = x_5 = 0$), for we would then have

$$|y| = (\sum y_i^2)^{1/2} < (\sum x_i^2)^{1/2} = \sigma$$

But this contradicts the fact that

$$|y| = |Px| = |x| = \sigma$$

We can, however, accomplish what we want by changing only *one* component (other than those we make zero). To see this, we again examine the illustration and suppose that we are willing to change the fourth component. We want, therefore, to get

$$P \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ 0 \\ y_4 \\ 0 \\ x_6 \end{pmatrix}$$

We have

$$x_1^2 + x_2^2 + y_4^2 + x_6^2 = \sigma^2$$

so therefore

$$\begin{aligned} y_4 &= \pm [\sigma^2 - (x_1^2 + x_2^2 + x_6^2)]^{1/2} \\ &= \pm [x_3^2 + x_4^2 + x_5^2] \end{aligned}$$

To find P we have

$$\begin{aligned} y &= Px = (I - 2vv^T)x \\ &= x - 2v(v^Tx) \end{aligned}$$

or

$$2(v^T x)v = x - y$$

where $v^T x$ is a scalar and v is a unit vector. The direction of v is obviously that of $x - y$. Thus

$$v = \frac{x - y}{|x - y|} = \frac{1}{|x - y|} \begin{pmatrix} 0 \\ 0 \\ x_3 \\ x_4 - y_4 \\ x_5 \\ 0 \end{pmatrix}$$

so we can choose u in the original definition of $P = I - uu^T/b$ as

$$u = \begin{pmatrix} 0 \\ 0 \\ x_3 \\ x_4 - y_4 \\ x_5 \\ 0 \end{pmatrix}$$

with y_4 given as above.

We have a choice of the sign for y_4 so we choose y_4 to have the opposite sign as x_4 , thus avoiding possible cancellation and roundoff. In practice, u is easier to use than v , because it avoids the square root necessary to find $|x - y|$.

Notice that y_4 and u depend only on the components of x that will be changed by the transformation. In doing the computation, one can ignore the components that are not changed.

The particular column (row) vectors used are the columns (rows) of A . When we form PA we, in effect, multiply each column of A by P , and when we form AP we multiply each row of A by P .

Returning to the particular six-dimensional case of our example, if

$$\begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} & a_{1,5} & a_{1,6} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} & a_{2,5} & a_{2,6} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} & a_{3,5} & a_{3,6} \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} & a_{4,5} & a_{4,6} \\ a_{5,1} & a_{5,2} & a_{5,3} & a_{5,4} & a_{5,5} & a_{5,6} \\ a_{6,1} & a_{6,2} & a_{6,3} & a_{6,4} & a_{6,5} & a_{6,6} \end{pmatrix}$$

and $a_{i,j} = a_{j,i}$ we can pick an α_1 , as described above, and set

$$u_1 = \begin{pmatrix} 0 \\ \alpha_1 \\ a_{3,1} \\ a_{4,1} \\ a_{5,1} \\ a_{6,1} \end{pmatrix}$$

Thus

$$P_1 = I - \frac{u_1 u_1^T}{b_1} \quad \text{with} \quad b_1 = \frac{|u_1|^2}{2}$$

so that PA has its first column equal to

$$\begin{pmatrix} a_{1,1} \\ a'_{2,1} \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

for some value of $a'_{2,1}$. As a result $(P_1 A)P_1$ will have its first row equal to

$$(a_{1,1}, a'_{1,2}, 0, 0, 0, 0)$$

with $a'_{1,2} = a'_{2,1}$. Setting $A_2 = P_1 A_1 P_1$ we again have a symmetric matrix similar to A .

$$A_2 = \begin{pmatrix} a_{1,1} & a'_{1,2} & 0 & 0 & 0 & 0 \\ a'_{2,1} & a'_{2,2} & \dots & \dots & \dots & a'_{2,6} \\ 0 & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & \dots & \dots & \dots & \dots \\ 0 & a'_{6,2} & \dots & \dots & \dots & a'_{6,6} \end{pmatrix}$$

If we now choose α_2 appropriately, we have

$$u_2 = \begin{pmatrix} 0 \\ 0 \\ \alpha_2 \\ a'_{4,2} \\ a'_{5,2} \\ a'_{6,2} \end{pmatrix}$$

and $P_2 = I - (u_2 u_2^T)/b_2$ as before, and $P_2 A_2$ will have its second column equal to

$$\begin{pmatrix} a'_{1,2} \\ a'_{2,2} \\ a''_{3,2} \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

P_2 will not change the first column or row of A_2 . Similarly, $(P_2 A_2)P_2$ will have the second row equal

$$(a'_{2,1}, a'_{2,2}, a''_{2,3}, 0, 0, 0)$$

with $a''_{2,3} = a''_{3,2}$. Then $A_3 = P_2 A_2 P_2$ will also be a symmetric matrix similar to A but with *two* columns and rows reduced to tridiagonal form.

Continuing in this way, we obtain a matrix B which is in tridiagonal form and similar to A . Further, to effect the similarity

$$B = Q A Q^{-1}$$

we have chosen Q as the product of the Householder matrices which are orthogonal. This implies that Q is also orthogonal, and

$$B = Q A Q^T$$

Note that in the case where A is not symmetric, we can still “zero out” the columns as we did above, but the same Householder transformations will no longer zero out the rows. We will obtain, in the nonsymmetric case, a matrix similar to A and of the form

$$\begin{pmatrix} \times & \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \\ 0 & 0 & \times & \times & \times & \times \\ 0 & 0 & 0 & 0 & \times & \times \end{pmatrix}$$

This is called the *Hessenberg* form.

It is worth noting that if we do not require that the final matrix be similar to A (so that the eigenvalues no longer are the same), then we need multiply A on the *left* only by a properly chosen sequence of Householder matrices and will end up with an upper triangular matrix. No requirement on the symmetry of the original matrix would then occur.

Because the Householder matrix is its own inverse, this remark is equivalent to writing the matrix A in the form

$$A = Q R.$$

where R is upper triangular and Q (being the product of Householder matrices) is orthogonal. This procedure has application not only in the “ QR algorithm” which we discuss in the next section, but also in the solution of systems of linear equations (Chap. 7), particularly the least-squares solution of redundant equations (Chap. 27).

45.7 THE QR ALGORITHM

The QR algorithm is used to compute the eigenvalues of a matrix. The basic idea is expressed above, namely, that we can start with a matrix A (call it A_1) and write it in the form

$$A_1 = Q_1 R_1$$

with Q_1 orthogonal and R_1 upper triangular. We then define

$$A_2 = R_1 Q_1$$

Note that

$$\begin{aligned} A_2 &= Q_1^{-1} A_1 Q_1 \\ &= Q_1^T A_1 Q_1 \end{aligned}$$

so that A_2 is similar to A_1 , and is symmetric if A_1 is symmetric. Continuing this process, we write

$$A_2 = Q_2 R_2$$

with Q_2 orthogonal and R_2 upper triangular, and define

$$A_3 = R_2 Q_2$$

In this way we obtain a sequence of matrices $A_1, A_2, A_3 \dots$, all similar to A . It is an important fact that if A is symmetric and tridiagonal, then so are all the other matrices of this sequence.

This observation is useful in computing the eigenvalues of a symmetric matrix. Given a symmetric matrix we first reduce it to tridiagonal form by Householder transformations and use this result as the starting matrix of the sequence A_i of similar matrices. In this latter part we need to compute only elements on the three main diagonals, and this results in a large savings in computation.

It has been shown by Francis¹ that for any matrix A , the matrices A_n tend “generally” to an upper triangular form.² In particular, for A symmetric the matrices A_n are symmetric and will therefore tend to diagonal form. The numbers along the diagonal will, of course, be the eigenvalues of A . Francis also introduced the operations called “shifts” to be performed at each step, whose purpose is to speed convergence. These shifts are useful but they do not change the general theory, and will not be discussed further here.

45.8 OVERDETERMINED SYSTEMS OF LINEAR EQUATIONS

Suppose that we are given a system of m linear equations in n unknowns with $m > n$,

$$Ax = y$$

where A is an $m \times n$ matrix, and x and y are, respectively, n - and m -dimensional column vectors. We write the equations in a symbolic block form as in Fig. 45.8.1

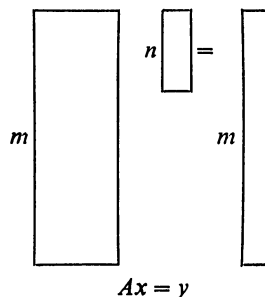


FIGURE 45.8.1

In general there will be no solution to such an overdetermined system of equations, but we may look for a least-squares solution as in Chaps. 25 to 27. This means that we want to find a vector x which minimizes the norm

$$|Ax - y|$$

¹ J. G. F. Francis, The QR Transformation, *Comp. J.* 4, pts. I and II, pp. 265–271, 332–345, 1961–1962.

² If A has nonreal eigenvalues, this statement must be modified because a triangular matrix has its eigenvalues on the diagonal while the process we described uses only real numbers. In the case in this chapter, when A is symmetric we have proved that the eigenvalues are all real.

Using the methods we have developed, we can find a sequence of Householder matrices whose product is an orthogonal matrix Q such that QA is in upper triangular form

$$QA = R = \begin{bmatrix} xxx & & \\ & xx & \\ & & x \\ 0 & & & \end{bmatrix}$$

We will assume—what is true most of the time—that the rank of the matrix is n (if not, then the details can be filled in) so that all the diagonal elements are nonzero. If we set $Qy = z$, then we have

$$|Rx - z| = |QAx - Qy| = |Q(Ax - y)| = |Ax - y|$$

since the orthogonal matrix Q does not change the norm. Thus we can equivalently minimize the quantity $|Rx - z|$.

Since R is upper triangular, we can write this equation symbolically in the form

$$R = \begin{bmatrix} \hat{R} & n \\ 0 & m - n \end{bmatrix} \quad Z = \begin{bmatrix} \hat{z} & n \\ z' & m - n \end{bmatrix}$$

Then

$$|Rx - z|^2 = |\hat{R}x - \hat{z}|^2 + |z'|^2$$

Now x is independent of $|z'|^2$, so to minimize $|Rx - z|$ we choose x to minimize $|\hat{R}x - \hat{z}|^2$. Since R is of rank n (by assumption), we can make the latter norm equal to zero by simply solving the triangular system

$$\hat{R}x = \hat{z}$$

exactly. The result is that this solution must be the least-squares solution to the original overdetermined system, and the norm of the residuals is given by $|z'|$.

N + 1

THE ART OF COMPUTING FOR SCIENTISTS AND ENGINEERS

N+1.1 IMPORTANCE OF THE TOPIC

It is unusual¹ in a book on computing to include a chapter on the vague general topic of how to approach and solve problems using computing machines. The title of the chapter is itself ambiguous; it could refer to the art as practiced by a person computing for scientists and engineers, or it could refer to the art as practiced by scientists and engineers; in fact, it is meant for both.

The subject should not be taken lightly just because at times it is opinion rather than established fact. I feel that it is more important than many of the specific results discussed in other parts of the book. At present, the subject is more an art than a science, but this situation is slowly changing because the presence of computers themselves has made possible the mechanization of many processes once believed to require human thought, and active research is now going on in this important area known popularly as artificial intelligence. The more we learn about how we solve problems, the more work we shall be able to shift from ourselves to the machines. The art of solving problems with the use of computing machines, is of interest in its own right; it can also help the user in many

¹ But see F. S. Acton [1].

situations and greatly increase the value of the machine computations now being done.

Most scientists seem to have been afraid to explore what is generally considered to be the creative process of discovery, but there have been some notable exceptions. Among the mathematicians, "The Method" by Archimedes is one of the earliest examples, and we now have the classic "How to Solve It" by Pólya [48]. Both, however, are concerned with the attack on well-formulated problems, whereas we are concerned with a larger framework of vaguely defined problems from which we hope to extract results to fit an equally vaguely defined situation. The motto of the book,

THE PURPOSE OF COMPUTING IS INSIGHT,
NOT NUMBERS

illustrates our broad frame of reference

Unfortunately, mathematics traditionally emphasizes elegance, which all too often is identified with surprise. Frequently in a mathematical text one will see *consider the following function . . .* and from the arguments that follow be compelled to agree with the result, without being given the slightest clue as to why we should consider the function or why it produces the result. Many mathematicians, including the famous C. F. Gauss, pride themselves on removing all the scaffolding that was used to erect the result in the first place and leave only the isolated result to be admired (surprised by). The motto is clearly in direct opposition to this attitude as it applies to numerical methods.

N + 1.2 WHAT ARE WE GOING TO DO WITH THE ANSWER?

Pólya in his book "How to Solve It" emphasizes the importance of understanding the problem. The present author, from many years' experience in computing for others, has found that usually the first question to ask is, What are we going to do with the answers? Will the answers computed actually answer the questions that should have been asked? Frequently one sees proposed studies, and even published papers, that cannot possibly answer the questions they claim will be (or are) answered simply because sufficient attention was not given to whether the data (and computed numbers) are or are not adequate to decide between the possible alternatives.¹

Again one may ask, Do we need all the answers? Do we need more? Would something else provide a better basis for insight? Many other questions also occur, such as, Are the known conservation laws obeyed by the results?

¹ See, for example, Jeffreys [23].

In order to answer some of these questions, it is well to imagine typical answer sheets and then examine these for their usefulness. More times than one would expect the requested answers will not answer the needs of the research project. The original request may have been to get the answers to a set of simultaneous equations. Sometimes this is all that the computing can give, but many times other items, such as the difficulty of solution, can add to the understanding of the situation being examined. Further, what is to be the measure of accuracy: accuracy in the unknowns, the residuals, or the change in the problem, to name but a few? Were the simultaneous equations necessary? Would an alternative formulation give more insight?

Before going on in this vein, let it also be observed that one cannot expect the problem proposer to know exactly what he wants. In research, and in many stages of development, it is in the nature of the process that we do not know exactly what we are seeking. Indeed, it may be said, "In research, if you know what you are doing, then you shouldn't be doing it." In a sense, if the answer turns out to be exactly what you expected, then you have learned nothing new, although you may have had your confidence increased somewhat.¹

Trite and obvious as the remark may be, it is important to know what you are seeking. It is less well understood that you should also plan your work to increase the chances that an unusual observation will be found. It is usually worth adding a small amount to the total machine time if in the process many side checks on the model being studied can be included. Thus, for example, in a study of the design of traveling wave tubes the author found that the equations being solved were a result of "linearizing the problem," and he therefore included in the computation some estimates of the probable size of the main nonlinear effects and found (not surprisingly) that in some cases for some ranges of the independent variable this was much too large to be ignored, thus invalidating the modeling, the corresponding computation, and nicely printed sheets of output.

Many of the greatest discoveries have been made by prepared minds making a chance observation and realizing the importance of it. Thus, even though it puts a little extra burden on the output equipment, it is well when planning the output formats to include a few wisely chosen numbers beyond the bare minimum.

In summary, while there are exceptions, it is a good general rule to begin a problem in computation with a searching examination of "What are we going to do with the answers?" An active, imaginative mind can make great contributions to the whole research problem at this stage, and a dull, lazy one can prevent any real insight from emerging from all the hours of machine time used to get the obvious numbers.

¹ P. Debye: "If a problem is clearly stated, it has no more interest to the physicist."

One of the most common mistakes in planning a problem is to request too many answers. This is particularly true in problems having many parameters. In such situations, what is generally needed is a good statistician who understands *the design of experiments and optimization theory*. Very often he can plan the search in such a fashion that only a small fraction of the original cases need be solved. The sheer volume of answers can, and very often does, stifle insight.

N+1.3 WHAT DO WE KNOW?

Having decided, tentatively, what we expect to get out of the computation, we next ask What do we know? What information do we have? What is the input? Are we putting in all that we know? For example, if the answer is known to go through the origin, have we included that fact in the input? Do we know some conservation law?

Again we remind the reader that Pólya emphasizes the importance of understanding the problem, but again he is considering mathematically formulated problems and is supposing that there is a complete statement. In the application of mathematics this is, of course, not the situation, nor is it true even in mathematical research. Often further probing of the situation being explored will bring further information to light. In the least-squares fitting of a polynomial in Sec. 27.6 we first ignored the fact that the curve should go through the origin, and as a result we had to do the problem over again. When we used this information, we found a more satisfactory answer.

Sometimes it is difficult to include all the known information in our formulation. Thus, in the above case we knew that the first term had a positive coefficient. We did not include this fact in the input but saved it for an output check. Nevertheless, it is well to get clearly in mind all the relevant information available before going on to the next stage.

Sometimes this exploration of the unknown situation will reveal alternative formulations of the problem, and these in turn may suggest new ideas to be explored. Sometimes we find that unnecessarily restrictive assumptions have been made on the model, and that with little or no extra effort they can be removed. Typically a problem has been needlessly linearized, or viscous damping has been removed. In any case, the role of assumptions should be understood, and checks that will throw light on the validity of some of the assumptions should be included in the computation. Thus the exploration of the input may reveal new or different output requirements.

N + 1.4 DESIGNING THE COMPUTATION ROUTINE

Only after we have clearly in mind where we are and where we want to be should we turn to serious work on the question, How do we get from here to there? This is the domain of Pólya's book; all his remarks are relevant, and we assume that the reader either is familiar with them or else will read the book.

Although a problem is posed as one suited to a computing machine, the necessity for using a computer should be questioned. An analytical answer is often very much superior to a numerical result, and sometimes, even when it is more difficult to compute than the original problem, the error estimates may be made much more accurately. Our exploration of what we know about a problem may produce one or more formulations of the problem, some of them merely simple mathematical transformations of the same problem and others, perhaps, quite different. It is expensive to explore a machine-computation plan for all of them, and some initial choices must be made. As a general rule (with many exceptions), the closer the mathematical statement to the fundamental concepts of the field the better, always assuming that scale transformations have been made to make the equations dimensionless. Fancy mathematical transformations usually introduce difficult computational situations.

The plan of computation adopted should make use of as much of the initial information as can be included conveniently. The various mathematical approximations made by the formulas used should be in harmony with the model adopted. The effects of sampling should be examined, as well as the effect of the limited amount of input data.

The plan of computation should include plans for checking both the coding and the results. This point is too often overlooked; thus we recommend, *before* doing any detailed programming, that the questions be asked, How shall I know if the answer is the one that is wanted? What tests shall I apply or have the machine apply? It is essential that some redundant information be computed or found from other sources, so that some checks can be made. It is the experience of the author that a good theoretician can account for almost anything produced, right or wrong, or at least he can waste a lot of his time worrying about whether it is right or wrong.

N + 1.5 ITERATION OF THE ABOVE STEPS

We have acted as if these three stages can be completely separated, when in fact they are often interrelated. Nevertheless, it is well to keep the three stages clearly in mind and iterate around them as new information at one stage sheds new light

on another. It is the author's experience that rushing into the stage of arranging the details of the computation is the most common mistake. This is especially true of the computer expert, because there he feels at home and can show his skill. But all his skill is wasted if he works on the wrong problem or produces numbers that do not answer the real questions.

It is, of course, difficult to be expert enough in a particular field to ask suitable questions about the importance of what is being computed as compared with some other things that might be computed instead. But there is an art in asking questions. Socrates claimed not that he knew truth but that he knew how to ask the proper questions of man and draw the truth from him. He spoke of himself as a midwife. And it is in somewhat the same way that the consulting computer expert must approach the man with a problem. In the final analysis, the man with the problem must make the choices and do the research, but well-chosen suggestions from the outsider can help to clarify the nature of the choices and aid in the decisions that must be made at many stages of the work.

N+1.6 A CODE OF ETHICS

Because I have recommended that an active, aggressive role be played by the numerical analyst and that he get into the design stages of the problem and stay in through the interpretation and decision stages, it seems necessary to also suggest some restraints on his role. The following could serve as commandments.

- 1 Thou shalt not *impose* your ideas, opinions, and prejudices on the customer, for he is the man with the problem, not you.
- 2 Thou shalt avoid all unnecessary fancy mathematics, computer jargon, and attempts to raise the "falutin' index" of the problem.
- 3 Thou shalt not come between the customer and the machine, but shall stand at one side ready and willing to give all the help and aid you can to both.

N+1.7 ESTIMATION OF THE EFFORT NEEDED TO SOLVE THE PROBLEM

In any reasonably mature science, it is necessary to make estimates as to what will happen *before* spending time and money. In a sense, the more mature the field, the more accurate the estimates. Judged by this standard, computing is in an elementary state. Often only the poorest estimates are available.

Some of the estimates that should be made are the following:

- 1 Will the roundoff errors be serious; if so, to what extent?
- 2 Will the interval spacing be adequate?
- 3 If there is an iteration process, how many iterations may reasonably be expected?
- 4 How much time will it take to code and debug?
- 5 How will the answers be checked to assure that they are right?
- 6 How much machine time will be required?
- 7 When will the answers finally be available?

A glance at these questions will convince the average computing expert that the present state of the art leaves much to be desired. This is no reason for not trying to make estimates that are as realistic as possible. Furthermore, proper organization of the computer facility can greatly improve the estimates made for questions 4 and 7. The provision of monitor systems, automatic coding systems, debugging systems, and a short turnaround time on the computer are musts if these two items are to be estimated accurately. Stability of the computer center is vital.

This book has tried to approach a number of these questions in various ways. In particular, Part III is believed to provide a basis for many estimates. However, many of the approaches in the book are merely tentative starts in the direction of making reliable estimates, and much remains to be done.

A person who aspires to become a computer expert in the consulting field should not dismiss these questions as unanswerable but should recognize that reasonably accurate estimates are the mark of a capable craftsman—estimates that are neither too optimistic nor too pessimistic.

N+1.8 LEARNING FROM CHANGES IN THE PLAN

It is almost inevitable that as the computation progresses, new information becomes available, and changes in the plan must be contemplated. But before adopting a change in plan, an effort should be made to understand why the wrong choice was made in the first place. Does the change shed any light on the model being used? Should we still try to get the same kind of answers? Does the change suggest new or different checks on the validity of the model? Can some new insights be obtained either from the failure or from a new plan?

A change in plan should not be hastily patched in but should receive the same careful thought and planning as went into the original plan. As remarked before, if things go as planned, not much can be learned—it is from the unexpected that the great new insights can occasionally arise. Thus a situation which forces

a change in plan should be regarded as an opportunity rather than a failure. Of course, if the change was due to carelessness or failure to think, then it should be taken as another example of the value of preliminary planning and should be charged to stupidity.

It is all too tempting, when involved in running a problem, to be rushed into making small changes without considering the consequences and the implications, especially if the results have been promised at a certain time. Yet haste at this time can undo much of the earlier careful work. It is well to remember that "the man should be the master, not the machine," and by proper organization of the computing facility and one's personal work habits much can be done to keep this clear.

N + 1.9 THE OPEN SHOP PHILOSOPHY

If we believe that *The Purpose of Computing Is Insight, Not Numbers*, then it follows that the man who is to get the insight must understand the computing. If he does not understand what is being done, he is very unlikely to derive much value from the computation. The bare numbers he can see, but their real meaning may be buried in the computation. The professional programmer and numerical analyst should not block his view of the problem and of the results of the computation.

Eddington has an illuminating story of a man who went fishing with a certain-sized net. When he found that the fish caught had a minimum size, he concluded that this was the minimum size of the fish in the sea; he made the mistake of not understanding how the fishing was done. And so it is with computing; what comes out depends on what goes in *and* on how it is processed. Without an understanding of the processes used, it is likely that effects due to the model used in the computing will be confused with effects of the model adopted by the user when he formulated the problem.

It has further been found that frequently the process of computing sheds great light on the model being computed. Computing is a tool that supplies numerical answers, but is also an intellectual tool for examining the world.

It is not likely that great physical insights will arise in the mind of a professional coder who routinely codes problems. If insights are to arise, and they are what we most want, then it follows that the man with problems must comprehend and follow the computing. This does not mean that he must do all the detailed work, but without a reasonably thorough understanding of what the computer is doing, it is unlikely that he can either arrange his work to get maximum benefit from the computer or achieve the insights which can and do result from properly arranged computations.